

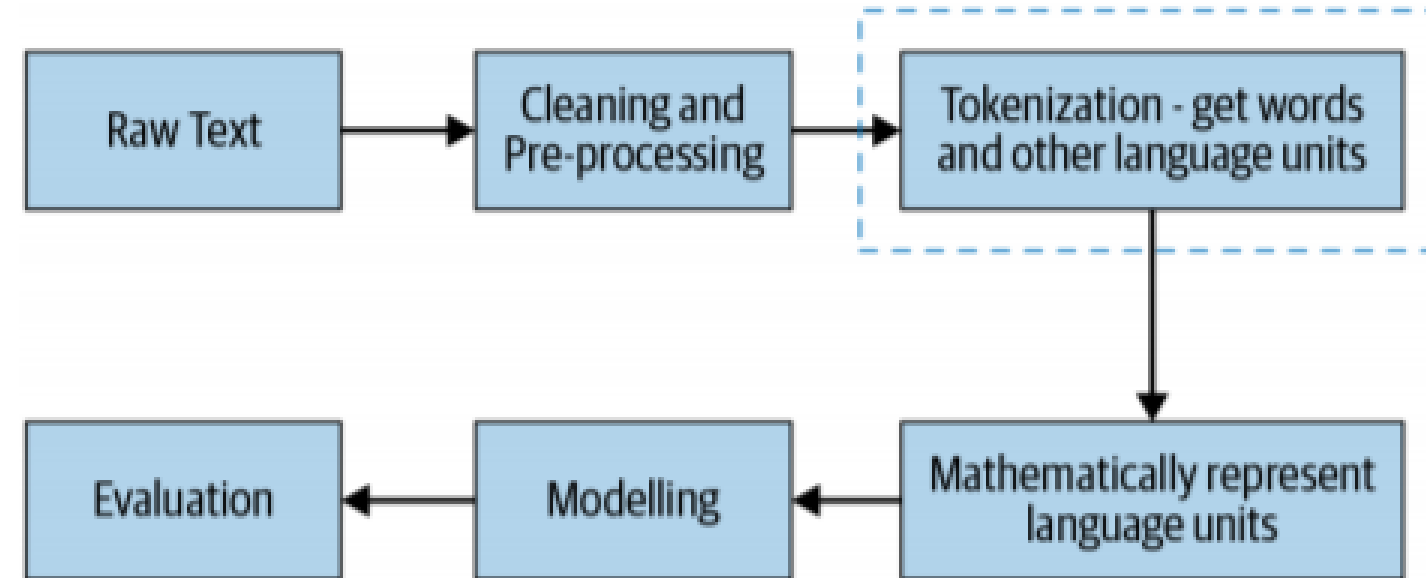
Applied NLP

DATA 641

Lecture 4 — Text Representation; Introduction

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Text Representation



How do we feed images to an ML model?

A good first approach is to consider the pixel values!



What We See

```
08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 91 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 62 00
81 49 31 73 55 79 14 29 93 71 40 67 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 69 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 45 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 67 10 26 38 40 67 59 54 70 66 18 38 64 70
67 26 20 68 02 62 12 20 95 63 94 39 63 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 96 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 56 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
19 80 81 68 05 94 47 69 28 73 92 13 86 52 17 77 04 89 55 40
04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
01 70 54 71 83 51 54 69 16 92 33 48 61 43 52 01 89 19 67 48
```

What Computers See

But what about text?

Typical approaches of text representation include

- Basic vectorization approaches
- Distributed representations
- Handcrafted features (More on Text Classification)

Why do we really need text representation?*

Let's say that we are given a labeled text corpus and asked to build a sentiment analysis model. To correctly predict the sentiment of a sentence, the model needs to understand the meaning of the sentence. In order to correctly extract the meaning of the sentence, the most crucial data points are:

- Break the sentence into lexical units such as lexemes, words, and phrases
- Derive the meaning for each of the lexical units
- Understand the syntactic (grammatical) structure of the sentence
- Understand the context in which the sentence appears

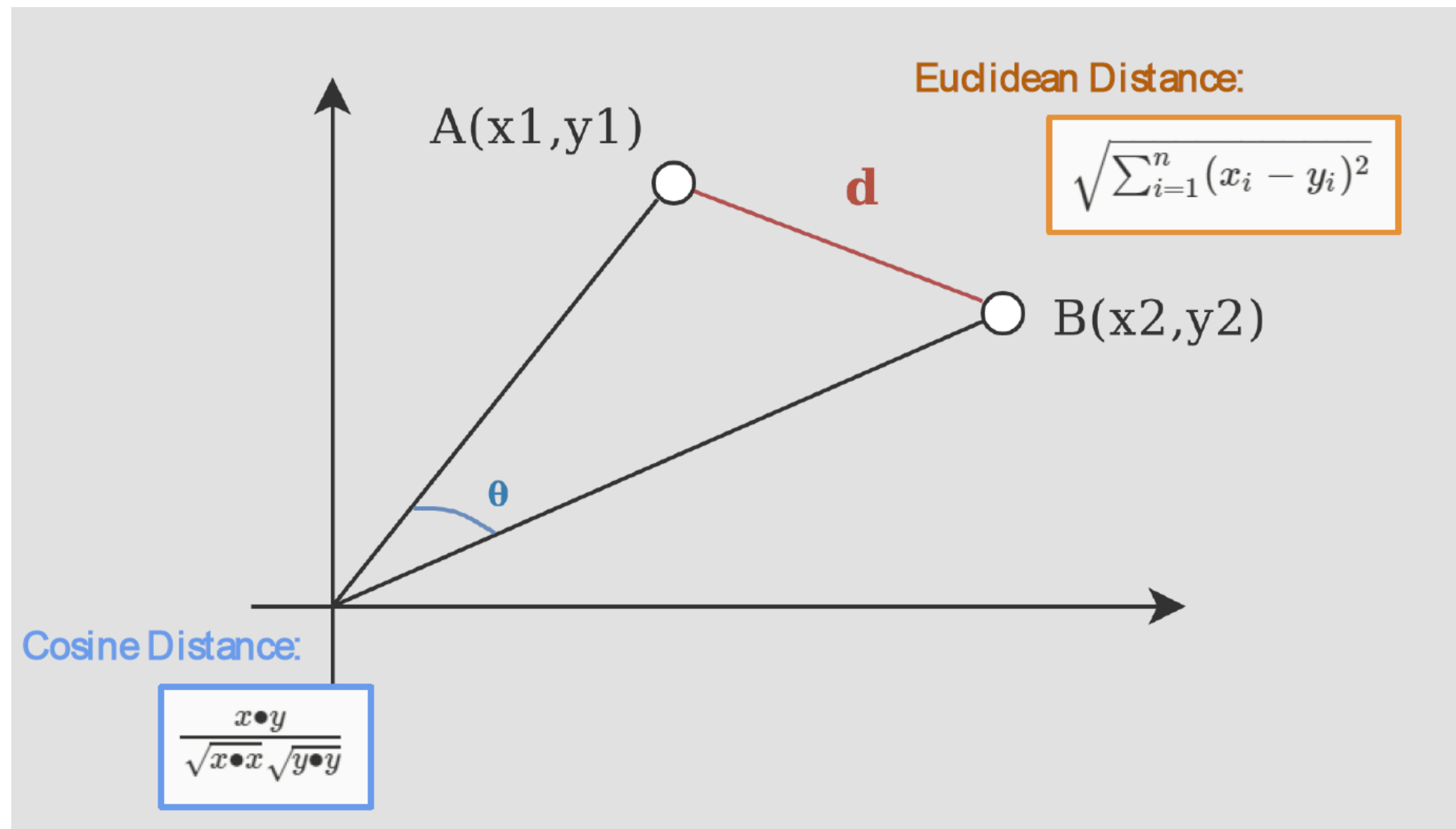
A good text representation scheme must facilitate the extraction of those data points in the best possible way to reflect the linguistic properties of the text. Without this, a text representation scheme is not of much use.

Similarity between text samples

Once we have the numerical representations of each text document how do we calculate how similar those two documents are?

- Need a function that compares two vectors, and returns a number.
- Measuring "similarity" in language can be quite complex. However, the most common way to calculate similarity between two text samples is to use cosine similarity.

Geometric interpretation



Euclidean vs Cosine Distance

- Euclidean distance:
 - Most "natural" sense of distance --i.e., the "straight-line" distance
 - Normalization could be an issue; discrepancies in magnitudes can cause huge distance values
- Cosine distance:
 - Looks at the "angle" formed between two vectors from origin to points. Recall $\cos(0) = 1.0$
 - Measure distance when the magnitude of the vectors does not matter. For instance text data represented by word counts